Fast Final Exponentiation on BW and BLS Curves with Even Embedding Degrees at 128 bits security

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Abstract

The final exponentiation is a crucial step in pairing computations, ensuring correctness and uniqueness of results in pairing-based cryptographic protocols. In this work, we propose an efficient method for computing the hard part of the final exponentiation on BW10-511, BW14-351 and BLS12 curves at 128 bits security level. Our approach reduces the computation cost by optimizing the exponentiation sequence and minimizing the number of required multiplications through an improved addition chain strategy. The computation cost of our method for the final exponentiation on these curves is about **25.6%**, **33.2%** and **10%** faster than the previously fastest result on BW10-511, BW14-351 and BLS12 curves respectively. The correctness of our formulas has been verified by a Magma code.

Keywords: Addition chain, Final exponentiation.

1 Introduction

A pairing is a non-degenerate bilinear map, $e : \mathbb{G}_1 \times \mathbb{G}_2 \longrightarrow \mathbb{G}_3$, where $\mathbb{G}_1$ and $\mathbb{G}_2$ are subgroups of an elliptic curve and $\mathbb{G}_3$ is a multiplicative sub-group of a finite field. Pairings are useful for designing several cryptographic schemes [1, 2]. The efficiency of these cryptographic schemes heavily depends on the speed of pairing computations. The implementation of pairings are based on two stages that are the Miller loop and the final exponentiation. Moreover, as the embedding degree increases, the cost of the final exponentiation surpasses that of the Miller loop. Therefore, improving the final exponentiation computation is a critical research focus. Many research have been made to improve the final exponentiation computation such as the vectorial addition-subtraction chain [3–5], the latticed-based method [6], and the method based on cyclotomic polynomials [7, 8] which is beneficial for the BLS family. The method via cyclotomic polynomials is the latest work focus on the final exponentiation that achieves a good performance on BW curves with prime embedding degree [9].

In 2023, the authors Yu Dai et *al.* [10] have revisited two pairings-friendly curves with even embedding degrees 10 and 14 over fields of size 511 and 351 respectively ensuring a security level of 128 bits called BW10-511 and BW14-351. Where for BW14-351 they provided high-speed software implementations of pairing computation, hashing to $\mathbb{G}_1$ and $\mathbb{G}_2$, group exponentiations, and subgroup membership testing on a 64-bit platform. Throughout their implementations, they deduced that BW14-351 is an appropriate choice for protocols that aims to pursue fast group exponentiations in $\mathbb{G}_1$ and $\mathbb{G}_T$, while minimizing the performance penalty for group exponentiations in $\mathbb{G}_2$. Furthermore, they proposed a fast algorithm to perform the final exponentiation step with lower cost multiplications on these curves. Otherwise, the authors Senegue Gomez et *al.* [11] extended the definition of the x-superoptimal pairing on BW10 and BW14 curves where their Miller loop is about 11.31% and 19.74% faster than the one of the optimal ate pairing done by Yu Dai et *al.* [10]. But they did not improved the final exponentiation step computation which make the pairing computation on these curves still less efficient and competitive. In addition, Razvan Barbulescu et *al.* [12] have proposed in their paper an up-to-date security evaluation for more than hundred pairing friendly elliptic curves. They have evaluated the complexity of a complete pairing execution taking into account the Miller algorithm for different degrees of twist and the final exponentiation for the most promising curves and found that the best pairings in the BD model [13] are BLS-24 and BLS-12 at 128 bits of security. The bilinearity of the x-superoptimal pairings and the correctness of our formulas with the new parameters have been verified by a Magma script available at https://github.com/N1960/gomez.git.

Our contribution. The contributions of this paper are as follows:

- We propose a new technique to reduce the size for the seed of BW10, BW14 and BLS12 curves without compromising the security of the curves which is useful to get a lower cost for the final exponentiation.
- Reducing the size of the seed makes the hard part exponent expansion very costly because of very large coefficients. We use a trick to save several multiplications and get an optimal addition chain.

- We compare the efficiency of our proposed final exponentiation computation on BW10, BW14 and BLS12 curves with the previously fastest result computations done by Yu Dai et *al.* [10] and Razvan Barbulescu et *al.* [12] respectively.

Roadmap. Section 2 gives a brief overview on pairing-friendly elliptic curves and addition chain. In section 3, we present new secure parameters for BW10, BW14 and BLS12 curves. In section 4, we evaluate the final exponentiation on BW10, BW14 and BLS12 curves. Section 5 summarizes operation costs of the previous sections and we end in section 6 with the conclusion.

2 Preliminaries.

In this section, we define a pairing-friendly elliptic curve [14] and an addition chain [15].

2.1 Pairing-friendly elliptic curve

Let $q(x), t(x), r(x) \in \mathbb{Q}[x]$ be non-zero polynomials. We say that a polynomial triple $(q(x), t(x), r(x))$ parametrizes a family of pairing-friendly ordinary elliptic curves with embedding degree k and CM discriminant D, if the following are satisfied :

(i) q(x) represents primes. That is, it is non-constant, irreductible, with positive leading coefficient. Additionally, $q(x) \in \mathbb{Z}$, for some (or infinitely many) $x \in \mathbb{Z}$ and $gcd(\{q(x) : x, q(x) \in \mathbb{Z}\}) = 1$.
(ii) r(x) is non-constant, irreducible, integer-valued, with positive leading coefficient.
(iii) r(x) divides both q(x)+1-t(x) and $\Phi_k(t(x) - 1)$, $\Phi_k(x)$ is the k^{th} cyclotomic polynomial.
(iv) there are infinitely many integer solutions (x, Y) for the parametrized CM equation $DY^2 = 4q(x) - t(x)^2$.

The ρ-value of a polynomial family $(q(x), t(x), r(x))$ is defined as $\rho(q, t, r) = \frac{degq}{degr}$. Let $f(x) = 4q(x) - t(x)^2 \in \mathbb{Q}[x]$ be the CM polynomial of the form $f(x) = g(x)y(x)^2$ with $y(x), g(x) \in \mathbb{Q}[x]$ and deg g ≤ 2. If deg g $= 0$, the family (q(x), t(x), r(x)) is complete and thus $f(x) = Dy(x)^2$, for some square-free $D > 0$. If deg g $= 1$, the family is complete with variable discriminant and finally, if deg g $= 2$, with g(x) not a perfect square and $lc(g) > 0$, the family is sparse.

2.2 Addition chain

Addition chains are useful to perform fast exponentiation in groups [15, 16] which is a crucial step for the computation of Tate pairing on elliptic curves. In this subsection, we define the concept of addition chain and the generalisation of an addition chain for a set of positive integers.

Definition 1 (Basic definition)**.** *An addition chain of a positive integer n is a sequence of positive integers* $\{e_0, e_1, e_2, ..., e_s\}$ *such that*

(i) $e_0 = 1, e_1 = 2$ *and* $e_s = n$.
(ii) for each i, $1 < i < s$, *there exist* $k, j < i$ *such that* $e_i = e_j + e_k$.

A generalization of the following definition take into consideration several integers.

Definition 2 (Generalization)**.** *A generalization of an addition chain of length l for a set of positive integers $\{n_1, n_2, ..., n_s\}$ is a sequence of positive integers $\{e_0, e_1, e_2, ..., e_l\}$ which include $\{n_1, n_2, ..., n_s\}$ such that*

(i) $e_0 = 1, e_1 = 2$ *and* $e_l = n_s$.

(ii) *for each* i, $1 < i < l$, *there exist* $0 \leq k \leq j < i$ *such that* $e_i = e_j + e_k$.

Such a chain defines a correct sequence of multiplications and squaring required for performing exponentiation on finite fields.

By means of Olivos theorem[17], group operations with the addition chain of length l can be accomplished with $l + s - 1$ operations in the group including squaring and multiplications of elements of the group. However, it has been shown that finding a minimal length addition sequence is an NP-hard problem.

3 New secure parameters of BW10-511, BW14-351 and BLS12 curves

In this section, we present the parameters $(T(x), q(x), r(x))$ obtained by Freeman et *al.* [14] for the curves BW10 and BW14 and Barreto et *al.* [18] for BLS12 curve. We also present a technique to get new secure parameters $(T(t), q(t), r(t))$ on these curves.

The parameters of the pairing-friendly elliptic curve BW10 are:

$$\begin{aligned} T(x) &= x^3 + 1 \\ r(x) &= x^8 + x^7 - x^5 - x^4 - x^3 + x + 1 \\ p(x) &= 1/3(x^3 - 1)^2(x^{10} - x^5 + 1) + x^3 \end{aligned}$$

For $x = 2^7 + 2^{13} + 2^{26} - 2^{32}$, Yu Dai et *al.* [10] found $p(x)$ and $r(x)$ primes of sizes 511 bits and 256 bits respectively corresponding to 128-bit security level with $p \equiv 1 \mod 3$ and they proposed a final exponentiation computation on BW10-511 curve.

Let's take $t = 1 + 2^6 + 2^{19} - 2^{25}$, then $x = 2^7 \times t$ since $x = 2^7 + 2^{13} + 2^{26} - 2^{32}$, so by replacing the expression of x on the parameters $(T(x), q(x), r(x))$, we obtain the new secure parameters $(T(t), q(t), r(t))$:

$$\begin{aligned} T(t) &= 2^{21}t^3 + 1 \\ r(t) &= 2^{56}t^8 + 2^{49}t^7 - 2^{35}t^5 - 2^{28}t^4 - 2^{21}t^3 + 2^7t + 1 \\ p(t) &= 1/3(2^{21}t^3 - 1)^2(2^{70}t^{10} - 2^{35}t^5 + 1) + 2^{21}t^3 \end{aligned}$$

Note that for $t = 1 + 2^6 + 2^{19} - 2^{25}$, the parameters $p(t)$ and $r(t)$ are still primes of sizes 511 bits and 256 bits respectively corresponding to 128-bit security level.

similarly, the parameters of the pairing-friendly elliptic curve BW14 are:

$$\begin{aligned}
T(x) &= x^8 - x + 1 \\
r(x) &= x^{12} + x^{11} - x^9 - x^8 + x^6 - x^4 - x^3 + x + 1 \\
p(x) &= 1/3(x-1)^2(x^{14} - x^7 + 1) + x^{15}
\end{aligned}$$

For $x = 2^6 - 2^{12} - 2^{14} - 2^{22}$, Yu Dai et *al.* [10] found $p(x)$ and $r(x)$ primes of sizes 351 bits and 265 bits respectively corresponding to 128-bit security level with $p \equiv 1 \mod 3$ and they proposed a final exponentiation computation on BW14-351 curve.

Let's take $t = 1 - 2^6 - 2^8 - 2^{16}$, then $x = 2^6 \times t$ since $x = 2^6 - 2^{12} - 2^{14} - 2^{22}$, so by replacing the expression of x on the parameters $(T(x), q(x), r(x))$, we obtain the new secure parameters $(T(t), q(t), r(t))$:

$$\begin{aligned}
T(t) &= 2^{48}t^8 - 2^6 t + 1 \\
r(t) &= 2^{72}t^{12} + 2^{66}t^{11} - 2^{54}t^9 - 2^{48}t^8 + 2^{36}t^6 - 2^{24}t^4 - 2^{18}t^3 + 2^6 t + 1 \\
p(t) &= 1/3(2^6 t - 1)^2(2^{84}t^{14} - 2^{42}t^7 + 1) + 2^{90}t^{15}
\end{aligned}$$

Note that for $t = 1 - 2^6 - 2^8 - 2^{16}$, the parameters $p(t)$ and $r(t)$ are still primes of sizes 351 bits and 265 bits respectively corresponding to 128-bit security level.

Furthermore, The parameters of the pairing-friendly elliptic curve BLS12 are:

$$\begin{aligned}
T(x) &= x + 1 \\
r(x) &= x^4 - x^2 + 1 \\
p(x) &= 1/3(x-1)^2 r(x) + x
\end{aligned}$$

For $x = -2^{77} + 2^{50} + 2^{33}$, Razvan Barbulescu et *al.* [13] found $p(x)$ and $r(x)$ primes of sizes 460 bits and 307 bits respectively corresponding to 128-bit security level and they proposed a final exponentiation computation on BLS12 curve.

Let's take $t = -2^{44} + 2^{17} + 1$, then $x = 2^{33} \times t$ since $x = -2^{77} + 2^{50} + 2^{33}$, so by replacing the expression of x on the parameters $(T(x), q(x), r(x))$, we obtain the new secure parameters $(T(t), q(t), r(t))$:

$$T(t) = 2^{33}t + 1$$
$$r(t) = 2^{132}t^4 - 2^{66}t^2 + 1$$
$$p(t) = 1/3(2^{33}t - 1)^2 r(t) + 2^{33}t$$

Note that for $t = -2^{44} + 2^{17} + 1$, the parameters $p(t)$ and $r(t)$ are still primes of sizes 460 bits and 307 bits respectively corresponding to 128-bit security level.

3.1 Arithmetic in $\mathbb{F}_{p^5}$, $\mathbb{F}_{p^7}$, $\mathbb{F}_{p^{10}}$, $\mathbb{F}_{p^{12}}$ and $\mathbb{F}_{p^{14}}$

In this section, we present the arithmetic cost in $\mathbb{F}_{p^5}$, $\mathbb{F}_{p^7}$, $\mathbb{F}_{p^{10}}$, $\mathbb{F}_{p^{12}}$ and $\mathbb{F}_{p^{14}}$.

Let M, S and I denote the cost of the multiplication, squaring and inversion in $\mathbb{F}_p$, whereas, M_k, S_k, S_c, cS_k, I_k, f_k^i, E_x denote the cost of the multiplication, squaring, cyclotomic squaring, compressed squaring, inversion, $p-th$ Frobenius operation and the power of x in F_{p^k} respectively.

In [19], the authors El Mrabet et *al* have reduced the multiplication in finite field extensions of degree 5 and 7 through the Newton's interpolation method where $M_5 = 9M + 137A_p$ and $M_7 = 13M + 271A_p$ and A_p is the cost of addition in $\mathbb{F}_p$.

So if we neglect the addition cost in $\mathbb{F}_p$, then we have the optimal cost in $\mathbb{F}_{p^5}$ and $\mathbb{F}_{p^7}$ [19].

Table 1 Operations cost by El Mrabet et *al.*

k	M_k	S_k	I_k	f_k^i
5	$9M$	$9M$	$65M$	$4M$
7	$13M$	$13M$	$102M$	$6M$

Where $I_5 = 3f_5^i + 2M_5 + I_1 + 10M$, $I_7 = 4f_7^i + 3M_7 + I_1 + 14M$, $I_1 = 25M$ and $f_k^i = (k-1)M$ if k is prime [20].

Also according to [10], Yu Dai et *al.* presented the following costs of arithmetic operations in the field $\mathbb{F}_{p^{10}}$ and $\mathbb{F}_{p^{14}}$ for the curves BW10-511 and BW14-351 respectively :

Note that in the table 2, the costs are obtained from [10] by neglecting some additions and modular reduction in $\mathbb{F}_{p^5}$ and $\mathbb{F}_{p^7}$, also by supposing that $M_u = M$ and $S_u = S$ in $\mathbb{F}_p$ where M_u and S_u represent the multiplication without reduction and the squaring without reduction in $\mathbb{F}_p$ respectively.

Table 2 Operation costs by Yu Dai et *al.*

k	M_k	S_k	S_c	I_k	f_k^i
10	$3M_5$	$2M_5$	$M_5 + S_5$	$I_5 + 2M_5 + 2S_5$	$8M$
14	$3M_7$	$2M_7$	$M_7 + S_7$	$I_7 + 2M_7 + 2S_7$	$12M$

Furthermore, the table 3 from [21] present the arithmetic on $\mathbb{F}_{p^{12}}$:

Table 3 arithmetic costs on $\mathbb{F}_{p^{12}}$.

Operation	$\mathbb{F}_{p^{12}}$
Multiplication	$54M$
Inversion	$119M$
Frobenius	$10M$
Squaring	$36M$
Cyclotomic Squaring (S_c)	$18M$
Compressed Squaring (cS_k)	$12M$
Simultaneous Decomposition of n elements	$(24n - 5)M + I_1$

4 Final exponentiation

For efficient computation of the final exponentiation, we use the method of Scott et *al.* [16] with the help of the addition chain. In this section we consider the embedding degree k to be even, therefore the expression $d(t) = \frac{p(t)^k - 1}{r(t)}$ can be written as :

$$\begin{aligned}\frac{p(t)^k - 1}{r(t)} &= \left(\frac{p(t)^k - 1}{\Phi_k(p(t))}\right) \times \left(\frac{\Phi_k(p(t))}{r(t)}\right)\\ &= ((p(t))^{k/2} - 1)\left(\frac{p(t)^{k/2} + 1}{\Phi_k(p(t))}\right) \times \left(\frac{\Phi_k(p(t))}{r(t)}\right).\end{aligned}$$

where $f^{((p(t))^{k/2}-1)\left(\frac{p(t)^{k/2}+1}{\Phi_k(p(t))}\right)}$ represent the easy part and $f^{\frac{\Phi_k(p(t))}{r(t)}}$ the hard part.

The exponentiation to the power of the easy part yields an element $f \in \mathbb{G}_{\Phi_k(p)}$ requiring only $3M_k + 1I_k + 2f_k^i$ for BW10 and BW14 curves and $2M_k + 1I_k + 2f_k^i$ for BLS12 curve. For efficiency, Scott et *al.* proposed to write the hard part $\frac{\Phi_k(p(t))}{r(t)}$ in base $p(t)$. In [6], based on the fact that a fixed power of a pairing is still a pairing, Fuentes et *al.* suggested to apply Scott et *al.*'s method with a power of any multiple d' of d with r not dividing d'. This could lead to a more efficient exponentiation as opposed to computing f^d directly.

4.1 The case of BW10

For the curve BW10 and the new parameters $p(t)$ and $r(t)$ define in section 3 :

We set $d'(t) = 3 \times d(t)$, so

$$3 \times \left(\frac{\Phi_{10}(p(t))}{r(t)}\right) = \sum_{i=0}^{3} m_i(t) p^i(t) \tag{1}$$

Where,

$$\begin{aligned}
m_0(t) = {} & 405648192073033408478945025720 32t^{15} - 316912650057057350374175801344t^{14} \\
& + 2475880078570760549798248448t^{13} - 38685626227668133590597632t^{12} \\
& + 302231454903657293676544t^{11} - 3541774862152233910272t^{10} \\
& + 18446744073709551616t^9 - 144115188075855872t^8 \\
& + 1688849860263936t^7 - 8796093022208t^6 + 68719476736t^5 - 268435456t^4 \\
& + 2097152t^3 + 32768t^2 - 384t + 3 \\
m_1(t) = {} & 316912650057057350374175801344t^{14} - 2475880078570760549798248448t^{13} \\
& + 19342813113834066795298816t^{12} - 453347182355485940514816t^{11} \\
& + 3541774862152233910272t^{10} - 27670116110564327424t^9 \\
& + 288230376151711744t^8 - 2251799813685248t^7 + 17592186044416t^6 \\
& - 103079215104t^5 + 805306368t^4 - 6291456t^3 + 16384t^2 + 256t - 2 \\
m_2(t) = {} & 151115727451828646838272t^{11} - 1180591620717411303424t^{10} \\
& + 9223372036854775808t^9 - 216172782113783808t^8 \\
& + 1688849860263936t^7 - 13194139533312t^6 + 103079215104t^5 \\
& - 805306368t^4 + 6291456t^3 - 16384t^2 + 128t - 1 \\
m_3(t) = {} & 72057594037927936t^8 - 562949953421312t^7 + 4398046511104t^6 \\
& - 68719476736t^5 + 536870912t^4 - 4194304t^3 + 16384t^2 - 128t + 1
\end{aligned}$$

When we extracts the absolute value of all the distinct coefficients of $m_i(t)$, we have a set of 40 elements given in powers numbers :

$$A = [1, 2, 3, 2^7, 2^8, 2^7 \cdot 3, 2^{14}, 2^{15}, 2^{21}, 2^{22}, 2^{21} \cdot 3, 2^{28}, 2^{29}, 2^{28} \cdot 3, 2^{36}, 2^{35} \cdot 3, 2^{42}, 2^{43}, 2^{42} \cdot 3, 2^{44}, 2^{49}, 2^{49} \cdot 3, 2^{51}, 2^{56}, 2^{57}, 2^{56} \cdot 3, 2^{58}, 2^{63}, 2^{64}, 2^{63} \cdot 3, 2^{70}, 2^{70} \cdot 3, 2^{77}, 2^{78}, 2^{77} \cdot 3, 2^{84}, 2^{85}, 2^{91}, 2^{98}, 2^{105}]$$

an optimal chain of the set B is of length $l = 68$ and it is given by :

$$\begin{aligned}B = [&1, \underline{2}, 3, \underline{2^2}, 2^4, \underline{2^5}, \underline{2^6}, \underline{2^7}, \underline{2^8}, 2^7 \cdot 3, \underline{2^9}, \underline{2^{10}}, \underline{2^{11}}, 2^{14}, \underline{2^{15}}, \underline{2^{16}}, \underline{2^{17}}, \underline{2^{18}}, \underline{2^{19}}, \underline{2^{20}}, \underline{2^{21}}, \underline{2^{22}},\\ &2^{21} \cdot 3, \underline{2^{23}}, \underline{2^{24}}, 2^{28}, \underline{2^{29}}, 2^{28} \cdot 3, \underline{2^{30}}, \underline{2^{31}}, 2^{34}, \underline{2^{35}}, \underline{2^{36}}, 2^{35} \cdot 3, 2^{38}, 2^{42}, \underline{2^{43}}, 2^{42} \cdot 3, \underline{2^{44}},\\ &\underline{2^{45}}, 2^{48}, \underline{2^{49}}, 2^{49} \cdot 3, \underline{2^{50}}, \underline{2^{51}}, \underline{2^{52}}, 2^{56}, \underline{2^{57}}, 2^{56} \cdot 3, \underline{2^{58}}, 2^{62}, \underline{2^{63}}, \underline{2^{64}}, 2^{63} \cdot 3, 2^{70}, \underline{2^{71}},\\ &2^{70} \cdot 3, 2^{76}, \underline{2^{77}}, \underline{2^{78}}, 2^{77} \cdot 3, 2^{84}, \underline{2^{85}}, 2^{90}, \underline{2^{91}}, 2^{98}, 2^{104}, \underline{2^{105}}]\end{aligned}$$

Remark 1. *To obtain an optimal chain for the set B, we use a trick that is for all* $n \in \mathbb{N}$, *we have* $2^n + 2^{n+1} = 2^n \times 3$, $2^n + 2^n \times 3 = 2^{n+2}$, $2^n + 2^n = 2^{n+1}$ *and* $2^{2n} = (2^n)^2$. *For example* $2^5 = 2^4 + 2^4$, $2^6 = (2^3)^2$, $2^{35} \times 3 = 2^{34} + 2^{35}$.

Otherwise, for an element $f \in \mathbb{G}_{\Phi_{10}(p)}$, we need to compute f^{x^i} $(1 \leq i \leq 15)$ and this cost $15(25S_{10} + 3M_{10}) = 15(25 \times 18 + 3 \times 27) = 7965M$ since we have $t = 1 + 2^6 + 2^{19} - 2^{25}$, where $\log_2(t) = 25$ and $HW(t) = 4$. Secondly, the Frobenius operator is applied to compute some of the 48 factors of the form $(f^{t^i})^{p^j}$ for $0 \leq i \leq 15$ and $1 \leq j \leq 3$ but in Eq. 1, only 36 such factors are required and this cost $36 \times 8 = 288M$.

Furthermore, in Eq. 1, the coefficient 1 appears twice which are in the equalities $m_2(t)$ and $m_3(t)$. Similarly, the coefficients 2 and 3 appear once, and the coefficients 128 appears twice . Those coefficients can be grouped as products of the factors $(f^{t^i})^{p^j}$. The variables y_0, y_1, y_2 and y_3 are used to group the coefficients 1, 2, 3 and 128 respectively as follows:

$$y_0 = \frac{1}{f^{p^2}} \cdot f^{p^3}, \quad y_1 = \frac{1}{f^p}, \quad y_2 = f, \quad y_3 = (f^t)^{p^2} \cdot \frac{1}{(f^t)^{p^3}}$$

Following this process, we can define a total of 40 coefficients y_k for $k = 0, ..., 39$. This leads us to the following multi-exponentiation problem,

$$f^{d'} = \prod_{k=0}^{39} y_k^{A[k]}$$

Where $A[k]$ takes values from the given list A.

The cost of extracting all the coefficients y_k from Eq. 1 is of 12 multiplications plus several negligible field inversions that can be computed by simply conjugation since the curve BW10 has an even embedding degree [22]. Finally, to solve this multi-exponentiation problem, we construct an addition chain from the absolute value of all the distinct coefficients that appear in the expressions $m_i(t)'s$ in Eq. 1. By Olivos theorem, the number of group operations is l + s - 1 = 68 + 40 - 1 = 107, which includes 41 doubling (numbers underlined) and 66 Multiplications in $\mathbb{F}_{q^{10}}$. Therefore this cost $66M_{10} + 41S_{10} + 12M_{10} = 66 \times 27 + 41 \times 18 + 12 \times 27 = 2844M$ and adding the calculation cost of the easy part $3M_{10} + 1I_{10} + 2f_{10}^i = 3 \times 27 + 1 \times 101 + 2 \times 8 = 198M$. Hence the final exponentiation cost is $7965M + 288M + 2826M + 198M = 11295M$.

4.2 The case of BW14

In this paragraph, we follow the same approach as done on BW10 curve to optimise the final exponentiation cost on BW14 curve.

For the curve BW14 and the new parameters $p(t)$ and $r(t)$ define in section 3 :

We set $d'(t) = 3 \times d(t)$, so

$$3 \times (\frac{\Phi_{14}(p(t))}{r(t)}) = \sum_{i=0}^{5} m_i(t) p^i(t) \tag{2}$$

Where,

$$\begin{aligned}
m_0(t) &= 1152921504606846976t^{10} + 281474976710656t^8 + 68719476736t^6 \\
&\quad + 12288t^2 - 192t + 3 \\
m_1(t) &= 1237940039285380274899124224t^{15} - 4722366482869645213696t^{12} \\
&\quad - 1152921504606846976t^{10} - 18014398509481984t^9 \\
&\quad - 562949953421312t^8 + 8796093022208t^7 - 274877906944t^6 \\
&\quad - 8192t^2 + 64t - 2 \\
m_2(t) &= 302231454903657293676544t^{13} + 4722366482869645213696t^{12} \\
&\quad + 73786976294838206464t^{11} + 1152921504606846976t^{10} \\
&\quad + 18014398509481984t^9 + 281474976710656t^8 + 4398046511104t^7 \\
&\quad + 3221225472t^5 - 16777216t^4 + 8192t^2 - 64t - 1 \\
m_3(t) &= -302231454903657293676544t^{13} - 4722366482869645213696t^{12} \\
&\quad - 73786976294838206464t^{11} - 1152921504606846976t^{10} \\
&\quad - 18014398509481984t^9 - 281474976710656t^8 - 2147483648t^5 \\
&\quad + 16777216t^4 + 262144t^3 - 8192t^2 + 64t + 1 \\
m_4(t) &= 4722366482869645213696t^{12} + 1152921504606846976t^{10} \\
&\quad + 281474976710656t^8 - 1073741824t^5 + 33554432t^4 \\
&\quad - 1048576t^3 + 8192t^2 - 64t - 1 \\
m_5(t) &= 16777216t^4 + 4096t^2 + 1
\end{aligned}$$

When we extracts the absolute value of all the distinct coefficients of $m_i(t)$, we have a set of 27 elements given in powers numbers :

$$C = [1, 2, 3, 2^6, 2^6 \cdot 3, 2^{12}, 2^{13}, 2^{12} \cdot 3, 2^{18}, 2^{20}, 2^{24}, 2^{25}, 2^{30}, 2^{31}, \\ 2^{30} \cdot 3, 2^{36}, 2^{38}, 2^{42}, 2^{43}, 2^{48}, 2^{49}, 2^{54}, 2^{60}, 2^{66}, 2^{72}, 2^{78}, 2^{90}]$$

an optimal chain of the set C is of length $l = 43$ and it is given by :

$$D = [1, \underline{2}, 3, \underline{2^2}, \underline{2^3}, 2^6, \underline{2^7}, 2^6 \cdot 3, \underline{2^8}, \underline{2^9} 2^{12}, \underline{2^{13}}, 2^{12} \cdot 3, 2^{14}, \underline{2^{15}}, 2^{18}, \underline{2^{19}}, \underline{2^{20}}, \underline{2^{21}}, 2^{24}, \underline{2^{25}}, \underline{2^{26}}, \\ \underline{2^{27}}, 2^{30}, \underline{2^{31}}, 2^{30} \cdot 3, \underline{2^{32}}, \underline{2^{33}}, 2^{36}, 2^{38}, \underline{2^{39}}, 2^{42}, \underline{2^{43}}, \underline{2^{44}}, \underline{2^{45}}, 2^{48}, \underline{2^{49}}, 2^{54}, \\ 2^{60}, 2^{66}, 2^{72}, 2^{78}, 2^{90}]$$

Note that, the optimal chain for the set C is obtained similarly like done on the set A using the remark 1.

Firstly, for an element $f \in \mathbb{G}_{\Phi_{14}(p)}$, we need to compute f^{t^i} $(1 \leq i \leq 15)$ and this cost $15(16S_{14} + 3M_{14}) = 15 \times 16 \times 26 + 15 \times 3 \times 39 = 7995M$ since we have $t = 1 - 2^6 - 2^8 - 2^{16}$ where $\log_2(t) = 16$ and $HW(t) = 4$. Secondly, the Frobenius operator is applied to compute some of the 80 factors of the form $(f^{t^i})^{p^j}$ for $(0 \leq i \leq 15$ and $1 \leq j \leq 5$ but in Eq. 2, only 46 factors are required and this cost $46 \times 12 = 552M$.

Furthermore, in Eq. 2, the coefficient 1 appears four times which are in the equalities $m_2(t)$, $m_3(t)$, $m_4(t)$ and $m_5(t)$. Similarly, the coefficient 2 appears once. Those coefficients can be grouped as products of the factors $(f^{t^i})^{p^j}$. The variables y_0 and y_1 are used to group the coefficients 1 and 2 respectively as follows:

$$y_0 = \frac{1}{f^{p^2}} \cdot f^{p^3} \cdot \frac{1}{f^{p^4}} \cdot f^{p^5}, \quad y_1 = \frac{1}{f^p}. \tag{3}$$

Following this process, we can define a total of 27 coefficients y_k for $k = 0, ..., 26$. This leads us to the following multi-exponentiation problem,

$$f^{d'} = \prod_{k=0}^{26} y_k^{C[k]}$$

Where $C[k]$ takes values from the given list C.

The cost of extracting all the coefficients y_k from Eq. 2 is of 25 multiplications plus several negligible field inversions that can be computed by simply conjugation since the curve BW14 has an even embedding degree [22]. Finally, to solve the multi-exponentiation problem, we construct an addition chain from the absolute value of all the distinct coefficients that appear in the expressions $m_i(t)'s$ in Eq. 2. By Olivos theorem, the number of group operations is l + s - 1 = 43 + 27 - 1 = $69M_{14}$, which includes 22 doubling (numbers underlined) and 47 Multiplications in $\mathbb{F}_{q^{14}}$. Therefore this cost $47M_{14} + 22S_{14} + 25M_{14} = 3380M$ and adding the calculation cost of the easy part $3M_{14} + 1I_{14} + 2f_{14}^i = 3 \times 39M + 1 \times 154M + 2 \times 12M = 295M$. Hence the final exponentiation cost is $7995M + 552M + 3380M + 295M = 12222M$.

4.3 The case of BLS12

In this paragraph, we follow the same process for the curve BLS12 as done for BW10 and BW14 curves. For the curve BLS12 and the new parameters $p(t)$ and $r(t)$ define in section 3 :

We set $d'(t) = 3 \times d(t)$, so

$$3 \times \left(\frac{\Phi_{12}(p(t))}{r(t)}\right) = \sum_{i=0}^{3} m_i(t) p^i(t) \tag{4}$$

Where,

$$\begin{aligned}
m_0(t) &= 46768052394588893382517914646921056628989841375232t^5 \\
&\quad - 1088903574147003083082798743781658276659 2t^4 \\
&\quad + 147573952589676412928t^2 - 8589934592t + 3 \\
m_1(t) &= 544451787073501541541399371890829138329 6t^4 \\
&\quad - 1267650600228229401496703205376t^3 \\
&\quad + 17179869184t - 1 \\
m_2(t) &= 63382530011411470074835160268 8t^3 \\
&\quad - 147573952589676412928t^2 \\
&\quad + 8589934592t \\
m_3(t) &= 73786976294838206464t^2 - 17179869184t + 1
\end{aligned}$$

When we extracts the absolute value of all the distinct coefficients of $m_i(t)$, we have a set of 11 elements given in powers numbers :

$$E = [1, 3, 2^{33}, 2^{34}, 2^{66}, 2^{67}, 2^{99}, 2^{100}, 2^{132}, 2^{133}, 2^{165}]$$

an optimal chain of the set E is of length $l = 30$ and it is given by :

$$\begin{aligned}
F = [&1, \underline{2}, 3, \underline{2^2}, 2^4, 2^8, \underline{2^9}, \underline{2^{10}}, \underline{2^{11}}, \underline{2^{12}}, 2^{16}, 2^{20}, 2^{24}, 2^{32}, \underline{2^{33}}, \underline{2^{34}}, \\
&2^{40}, \underline{2^{41}}, 2^{48}, \underline{2^{49}}, 2^{66}, \underline{2^{67}}, 2^{82}, 2^{98}, \underline{2^{99}}, \underline{2^{100}}, 2^{132}, \underline{2^{133}}, 2^{164}, \underline{2^{165}}]
\end{aligned}$$

Note that, the optimal chain for the set E is obtained similarly like done on the set A using the remark 1.

Otherwise, for an element $f \in \mathbb{G}_{\Phi_{12}(p)}$, we need to compute f^{x^i} $(1 \leq i \leq 5)$ and this cost $5(44\, cS_{12}+3M_{12}) = 5(44\times12+3\times54) = 3450M$ since we use the compressed squaring (cS_k) and we have $t = -2^{44}+2^{17}+1$, where $\log_2(t) = 25$ and $HW(t) = 3$, plus a decompression of 3 field elements equals to $(24\times3-5)M + I_1 = 67M + I_1$. Secondly, the Frobenius operator is applied to compute some of the 18 factors of the form $(f^{t^i})^{p^j}$ for $0 \leq i \leq 5$ and $1 \leq j \leq 3$ but in Eq. 4, only 10 such factors are required and this cost $10\times10 = 100M$.

Following the same process like in Eq. 3, we can define a total of 11 coefficients y_k for $k = 0, ..., 10$. This leads us to the following multi-exponentiation problem,

$$f^{d'} = \prod_{k=0}^{10} y_k^{E[k]}$$

Where $E[k]$ takes values from the given list E.

The cost of extracting all the coefficients y_k from Eq. 4 is of 4 multiplications plus several negligible field inversions that can be computed by simply conjugation since the curve BLS12 has an even embedding degree [22]. Finally, to solve this multi-exponentiation problem, we construct an addition chain from the absolute value of all the distinct coefficients that appear in the expressions $m_i(t)'s$ in Eq. 4. By Olivos theorem, the number of group operations is l + s - 1 = 30 + 11 - 1 = 40, which includes 15 doubling (numbers underlined) and 25 Multiplications in $\mathbb{F}_{q^{12}}$. Therefore this cost $25M_{12} + 15S_c + 4M_{12} = 1836M$ and adding the calculation cost of the easy part $2M_{12} + 1I_{12} + 2f_{12}^i = 247$. Hence the final exponentiation cost is $3450M + 67M + I_1 + 100M + 1836M + 247M = 5575M + 6I_1$.

5 Comparison

In Table 4, we compare the theoretical costs of the final exponentiation done by Yu Dai et *al.* [10] and Razvan Barbulescu et *al.* [12] with our work done in this paper for the BW10-511, BW14-351 and BLS12 curves.

Table 4 Comparison of the computation cost of the final exponentiation of our work with the final exponentiation done by Yu Dai et *al.* [10] and Razvan Barbulescu et *al.* [12] on BW10-511, BW14-351 and BLS12 curves respectively at 128-bit security level.

Curve	Pairing	Miller loop	Final exponentiation	Total cost
BW10-511	x-superoptimal [5, 10]	$9622M$	$15158M + I_1$	$24780M + I_1$
	x-superoptimal(this work)	$9622M$	$11270M + I_1$	$20892M + I_1$
BW14-351	x-superoptimal [5, 10]	$9944M$	$18260M + I_1$	$28204M + I_1$
	x-superoptimal (this work)	$9944M$	$12197M + I_1$	$22141M + I_1$
BLS12	Optimal Ate [12]	$7438M$	$6188M + 6I_1$	$13626M + 6I_1$
	Optimal Ate (this work)	$7438M$	$5575M + 6I_1$	$13013M + 6I_1$

Note that in the Table 4, the costs of the final exponentiation for BW10-511 and BW14-351 [10] curves is obtained by neglecting some additions and modular reduction in $\mathbb{F}_{p^{10}}$ and $\mathbb{F}_{p^{14}}$, also by supposing that $M_u = S_u = M$ in $\mathbb{F}_p$ where M_u and S_u represent the multiplication without reduction and the squaring without reduction in $\mathbb{F}_p$ respectively.

From Table 4, we observe that the final exponentiation of our new approach is about 25.6%, 33.2% faster than the one done by Yu Dai et *al.* [10] and 10% faster than the one done by Razvan Barbulescu et *al.* [12]. The overall improvement (Miller loop and final exponentiation) is about 15.7% and 21.5% for the x-superoptimal pairing on BW10 and BW14 curves respectively and 4.5% for the optimal ate pairing on BLS12 curves.

6 Conclusion

In this paper, we proposed an efficient method for computing the hard part of the final exponentiation on BW10-511, BW14-351 and BLS12 curves at 128 bits security level. The computation cost of our method for the final exponentiation on these curves is about 25.6%, 33.2% and 10% faster than the previously fastest result on BW10-511, BW14-351 and BLS12 curves respectively. The overall improvement (Miller loop and final exponentiation) is about 15.7% and 21.5% for the x-superoptimal pairing on BW10 and BW14 curves respectively and 4.5% for the optimal ate pairing on BLS12 curves. We also implemented the x-superoptimal pairing on BW10-511 and BW14-351 curves with the new parameters in the MAGMA software to ensure correctness of our formulas. As one of the future works, the authors would like to obtain similar results for the other families of curves.

Declaration

Author Contributions

All the authors conceived the study. Senegue Gomez Nyamsi has edited the manuscript and all authors reviewed the manuscript. The final version of the manuscript was vetted and approved by all authors.

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